Switch Hardware

11.3.1

Switch Platforms

One simple way to create hierarchical and scalable networks is to use the right equipment for the job. There is a variety of switch platforms, form factors, and other features that you should consider before choosing a switch.

When designing a network, it is important to select the proper hardware to meet current network requirements, as well as to allow for network growth. Within an enterprise network, both switches and routers play a critical role in network communication.

Click each button for more information about the categories of switches for enterprise networks.

- Campus LAN Switches
- Cloud-Managed Switches
- Data Center Switches
- Service Provider Switches
- Virtual Networking

Campus LAN Switches

To scale network performance in an enterprise LAN, there are core, distribution, access, and compact switches. These switch platforms vary from fanless switches with eight fixed ports to 13-blade switches supporting hundreds of ports. Campus LAN switch platforms include the Cisco 2960, 3560, 3650, 3850, 4500, 6500, and 6800 Series.

11.3.2

Switch Form Factors

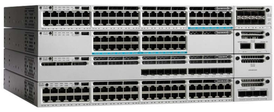
When selecting switches, network administrators must determine the switch form factors. This includes fixed configuration, modular configuration, stackable, or non-stackable

Click each button for more information about switch form factors.

- Fixed configuration switches
- Modular configuration switches
- Stackable configuration switches
- Thickness

Fixed configuration switches

Features and options on fixed configuration switches are limited to those that originally come with the switch.



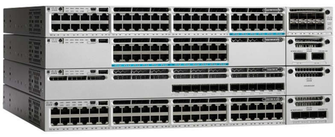
11.3.3

Port Density

The port density of a switch refers to the number of ports available on a single switch. The figure shows the port density of three different switches.

Fixed configuration switches support a variety of port density configurations. The Cisco Catalyst 3850 come in 12, 24, 48 port configurations, as shown in the figure. The 48-port switch has an option for additional ports for small form-factor pluggable (SFP) devices.

Cisco Catalyst 3850 Switches



Modular switches can support very high port densities through the addition of multiple switchport line cards. The modular Catalyst 9400 switch shown in the next figure supports 384 switchport interfaces.

Catalyst 9400 Switch

Large networks that support many thousands of network devices require high density, modular switches to make the best use of space and power. Without using a high-density modular switch, the network would need many fixed configuration switches to accommodate the number of devices that need network access. This approach can consume many power outlets and a lot of closet space.

The network designer must also consider the issue of uplink bottlenecks. A series of fixed configuration switches may consume many additional ports for bandwidth aggregation between switches, for the purpose of achieving target performance. With a single modular switch, bandwidth aggregation is less of an issue because the backplane of the chassis can provide the necessary bandwidth to accommodate the devices connected to the switchport line cards.

11.3.4

Forwarding Rates

Forwarding rates define the processing capabilities of a switch by rating how much data the switch can process per second. Switch product lines are classified by forwarding rates. Entry-level switches have lower forwarding rates than enterprise-level switches. Forwarding rates are important to consider when selecting a switch. If the switch forwarding rate is too low, it cannot accommodate full wire-speed communication across all of its switch ports. Wire speed is the data rate that each Ethernet port on the switch is capable of attaining. Data rates can be 100 Mbps, 1 Gbps, 10 Gbps, or 100 Gbps.

For example, a typical 48-port gigabit switch operating at full wire speed generates 48 Gbps of traffic. If the switch only supports a forwarding rate of 32 Gbps, it cannot run at full wire speed across all ports simultaneously. Fortunately, access layer switches typically do not need to operate at full wire speed, because they are physically limited by their uplinks to the distribution layer. This means that less expensive, lower performing switches can be used at the access layer, and more expensive, higher performing switches can be used at the distribution and core layers, where the forwarding rate has a greater impact on network performance.

11.3.5

Power over Ethernet

Power over Ethernet (PoE) allows the switch to deliver power to a device over the existing Ethernet cabling. This feature can be used by IP phones and some wireless access points, allowing them to be installed anywhere that there is an Ethernet cable. A network administrator should ensure that the PoE features are actually required for a given installation, because switches that support PoE are expensive.

Click each button to view PoE ports on different devices.

- Switch
- IP Phone
- WAP
- Cisco Catalyst 2960-C

Switch

PoE ports look the same as any other switch port. Check the model of the switch to determine if the port supports PoE.

11.3.6

Multilayer Switching

Multilayer switches are typically deployed in the core and distribution layers of an organization's switched network. Multilayer switches are characterized by their ability to build a routing table, support a few routing protocols, and forward IP packets at a rate close to that of Layer 2 forwarding. Multilayer switches often support specialized hardware, such as application-specific integrated circuits (ASICs). ASICs along with dedicated software data structures can streamline the forwarding of IP packets independent of the CPU.

There is a trend in networking toward a pure Layer 3 switched environment. When switches were first used in networks, none of them supported routing. Now, almost all switches support routing. It is likely that soon all switches will incorporate a route processor because the cost of doing so is decreasing relative to other constraints.

The figure shows a Catalyst 2960. Catalyst 2960 switches illustrate the migration to a pure Layer 3 environment. With IOS versions prior to 15.x, these switches supported only one active switched virtual interface (SVI). With IOS 15.x, these switches now support multiple active SVIs. This means that the switch can be remotely accessed via multiple IP addresses on distinct networks.

11.3.7

Business Considerations for Switch Selection

The following table highlights other common business considerations when selecting switch equipment.

Consideration	Description
Cost	The cost of a switch will depend on the number and speed of the interfaces, supported features, and expansion capability.
Port density	Network switches must support the appropriate number of devices on the network.
Power	It is now common to power access points, IP phones, and compact switches over Power over Ethernet (PoE). In addition to PoE considerations, some chassis-based switches support redundant power supplies.
Reliability	The switch should provide continuous access to the network.
Port speed	The speed of the network connection is of primary concern to end users.
Frame buffers	The ability of the switch to store frames is important in a network where there may be congested ports to servers or other areas of the network.
Scalability	The number of users on a network typically grows over time; therefore, the switch should provide the opportunity for growth.

11.3.8

Check Your Understanding – Switch Hardware

Check your understanding of switch hardware by choosing the BEST answer to the following questions.

1. Which category of switch is used to aggregate traffic at the edge of the network?
   - campus LAN switches
   - cloud-managed switches
   - service provider switches
   - data center switches

2. Field replaceable line cards are a feature found on which type of switch?
   - modular
   - fixed configuration
   - virtual
   - stackable

3. This is the term used to describe switches that can be interconnected and managed as one larger single switch.
   - modular
   - stackable
   - cloud-managed
   - multilayer

4. This is the term used to describe the number of ports available on a single switch.
   - speed
   - scalability
   - density
   - modularity

5. This is the term used to describe how much data a switch can process per second.
   - forwarding rate
   - power
   - reliability
   - wire speed

6. Which type of switch supports routing protocols and can forward IP packets at close to Layer 2 forwarding rates?
   - modular
   - stackable
   - cloud-managed
   - multilayer

Check

Show Me

Reset